20261020 The Origin of Fading Emission Lines and Increasing BPT Line Ratios along Infall Stages

1. Physical picture and disclaimer

We consider a local region in which RPS removes atomic gas but does not directly remove molecular gas. The loss of HI reduces the subsequent molecular supply. The retained H_2 reservoir then supports star formation while it is gradually consumed. Young-star Hα emission follows this declining SFR. When old stars and absorbing gas remain, hot low-mass evolved stars (HOLMES) can provide a more slowly varying ionizing contribution, whose fraction of the total Balmer emission increases. A forbidden-line-to-Balmer ratio rises if the HOLMES-powered emitting component has the larger intrinsic ratio.

For simplicity, we set up a restricted spatially resolved analytical model. Its conversion time, molecular depletion time, recycling fraction, feedback loading, and HI stripping coefficient are constant in time at a specified location. The Hα budget combines compact HII emission and leaked-OB-powered emission into one effective young-star component, with a constant HOLMES component and adopts the same intrinsic Hα/Hβ ratio, 2.86, for both components. Time $t = 0$ denotes the onset of the imposed environmental and supply conditions. It is not an observed time assigned to an infall-stage category.

2. Definition of the local two-reservoir model

2.1 Key quantities

Following the resolved regulator notation in Huang et al. (2026), section 4, we define SFR surface density and molecular GSR surface density as

$$\Sigma_{\rm SFR}(t) \equiv \frac{\Sigma_{\rm H_2}(t)}{\tau_{\rm dep}}, \qquad \Sigma_{\Phi}(t) \equiv \frac{\Sigma_{\rm H_2}(t)}{\tau_{\Phi}} \equiv \frac{\Sigma_{\rm HI}(t)}{\tau_{\rm conv}}. \tag{1}$$

Here $\tau_{\rm dep}$ is the molecular depletion time defined relative to the total rate of star formation. The gas supply-rate surface density $\Sigma_{\Phi}(t)$ feeds the molecular reservoir. The conversion time $\tau_{\rm conv}$ describes the assumed net effective transfer from the local atomic phase. The second equality is acting as the bridge to connect molecular and actomic

phase of Hydrogen gas in our model. Thus, the replenishment timescale of H_2 or consumption timescale of HI is

$$\tau_\Phi(t) \equiv \frac{\Sigma_{H_2}(t)}{\Sigma_\Phi(t)} = \tau_{conv}\frac{\Sigma_{H_2}(t)}{\Sigma_{HI}(t)}. \tag{2}$$

Consequently, $\Sigma_\Phi(t)$ is a rate, whereas τ_Φ and τ_{conv} are times. Even when τ_{conv} is constant, τ_Φ generally evolves.

Let $0 \leq R < 1$ be the prompt stellar mass return fraction and $\eta \geq 0$ the feedback mass-loading factor, so that the feedback mass loss rate is $\eta\Sigma_{SFR}(t)$. The net removal associated with star formation and feedback is $(1 - R + \eta)\Sigma_{SFR}(t)$. This is the usual regulator bookkeeping, here assigned effectively to the molecular reservoir (e.g., Lilly et al. 2013; Armitage 2026; Huang et al. 2026).

We use only γ_{strip} for the direct RPS loss coefficient and it has units of inverse time and acts only on HI gas. The baseline has no direct molecular stripping, and no external supply to HI after $t = 0$ due to the fact of starvation in cluster environment. Again, τ_{conv}, τ_{dep}, R, η, and γ_{strip} are constant in time at fixed location $\boldsymbol{x}$. Their possible spatial dependence is retained conceptually. Constant τ_{dep} means constant molecular SFE, not constant SFR. A linear molecular law is an empirical first approximation in nearby discs, with substantial environmental and scale-dependent limitations (Leroy et al. 2013). The HI-only stripping choice is a hypothesis for this calculation, not a claim that molecular stripping never occurs; observations provide counterexamples (Boselli et al. 2014).

2.2 Continuity equations

Removing each mass flux from the reservoir that supplies it gives

$$\begin{aligned} \frac{d\Sigma_{HI}(t)}{dt} &= -\Sigma_\Phi(t) - \gamma_{strip}\Sigma_{HI}(t), \\ \frac{d\Sigma_{H_2}(t)}{dt} &= \Sigma_\Phi(t) - (1 - R + \eta)\Sigma_{SFR}(t). \end{aligned} \tag{3}$$

These are the model's local mass balances. Their sum explicitly cancels the internal transfer:

$$\frac{d}{dt}(\Sigma_{HI}(t) + \Sigma_{H_2}(t)) = -\gamma_{strip}\Sigma_{HI}(t) - (1 - R + \eta)\Sigma_{SFR}(t). \tag{4}$$

Thus conversion does not destroy gas. The total cold reservoir declines through stripping and net stellar/feedback consumption. Furthermore, we define two physically labelled response rates,

$$\gamma_{\rm HI} \equiv \frac{1}{\tau_{\rm conv}} + \gamma_{\rm strip}, \qquad \gamma_{\rm H_2} \equiv \frac{1 - R + \eta}{\tau_{\rm dep}}. \tag{5}$$

The first is the total fractional removal rate from the HI reservoir, including conversion. The second is the net molecular consumption rate. Both use the same rate notation; $\gamma_{\rm strip}$ remains the only direct stripping coefficient.

Thus, we can rewrite the equation (3) as

$$\begin{aligned} \frac{d\Sigma_{\rm HI}(t)}{dt} &= -\gamma_{\rm HI}\Sigma_{\rm HI}(t), \\ \frac{d\Sigma_{\rm H_2}(t)}{dt} &= \Sigma_{\Phi}(t) - \gamma_{\rm H_2}\Sigma_{\rm H_2}(t). \end{aligned} \tag{6}$$

3. Explicit solution and overall SF suppression

3.1 Solve the atomic reservoir

For positive initial column $\Sigma_{\rm HI,0}$, the exact solution of atomic gas mass is

$$\Sigma_{\rm HI}(t) = \Sigma_{\rm HI,0} e^{-\gamma_{\rm HI} t}. \tag{7}$$

Immediately, we also have

$$\Sigma_{\Phi}(t) = \frac{\Sigma_{\rm HI,0}}{\tau_{\rm conv}} e^{-\gamma_{\rm HI} t}. \tag{8}$$

3.2 Molecular reservoir and explicit SFR solution

First, insert equation (7) and (8) into the second balance:

$$\frac{d\Sigma_{\rm H_2}(t)}{dt} + \gamma_{\rm H_2}\Sigma_{\rm H_2}(t) = \frac{\Sigma_{\rm HI,0}}{\tau_{\rm conv}} e^{-\gamma_{\rm HI} t}. \tag{9}$$

Multiplication by $e^{\gamma_{\rm H_2} t}$ makes the left-hand side a product derivative:

$$\begin{aligned} \frac{d}{dt}\left[e^{\gamma_{\rm H_2} t}\Sigma_{\rm H_2}(t)\right] &= e^{\gamma_{\rm H_2} t}\frac{d\Sigma_{\rm H_2}(t)}{dt} + \gamma_{\rm H_2} e^{\gamma_{\rm H_2} t}\Sigma_{\rm H_2}(t), \\ &= \frac{\Sigma_{\rm HI,0}}{\tau_{\rm conv}} e^{(\gamma_{\rm H_2} - \gamma_{\rm HI}) t}. \end{aligned} \tag{10}$$

Integrating the product derivative from 0 to t gives

$$e^{\gamma_{H_2} t}\Sigma_{H_2}(t) - \Sigma_{H_2,0} = \frac{\Sigma_{HI,0}}{\tau_{conv}} \int_0^t e^{(\gamma_{H_2}-\gamma_{HI})u}\, du. \tag{11}$$

For $\gamma_{HI} \neq \gamma_{H_2}$, the integral is

$$\begin{aligned} \int_0^t e^{(\gamma_{H_2}-\gamma_{HI})u}\, du &= \left[\frac{e^{(\gamma_{H_2}-\gamma_{HI})u}}{\gamma_{H_2}-\gamma_{HI}}\right]_0^t \\ &= \frac{e^{(\gamma_{H_2}-\gamma_{HI})t} - 1}{\gamma_{H_2}-\gamma_{HI}}. \end{aligned} \tag{12}$$

Now add the initial column, multiply by $e^{-\gamma_{H_2} t}$, and distribute the exponential:

$$\begin{aligned} \Sigma_{H_2}(t) &= e^{-\gamma_{H_2} t}\left[\Sigma_{H_2,0} + \frac{\Sigma_{HI,0}}{\tau_{conv}} \frac{e^{(\gamma_{H_2}-\gamma_{HI})t} - 1}{\gamma_{H_2}-\gamma_{HI}}\right] \\ &= \Sigma_{H_2,0} e^{-\gamma_{H_2} t} + \frac{\Sigma_{HI,0}}{\tau_{conv}} \frac{e^{-\gamma_{HI} t} - e^{-\gamma_{H_2} t}}{\gamma_{H_2}-\gamma_{HI}}. \end{aligned} \tag{13}$$

The first term is the evolution of surviving initial molecular gas. The second is the evolution of molecular gas supplied after $t = 0$, with its subsequent consumption included.

Finally, dividing every term by the constant depletion time gives the explicit SFR solution:

$$\boxed{\begin{aligned} \Sigma_{SFR}(t) = &\frac{\Sigma_{H_2,0}}{\tau_{dep}} e^{-\gamma_{H_2} t} \\ &+ \frac{\Sigma_{HI,0}}{\tau_{conv}\tau_{dep}} \frac{e^{-\gamma_{HI} t} - e^{-\gamma_{H_2} t}}{\gamma_{H_2}-\gamma_{HI}}. \end{aligned}} \tag{14}$$

If the HI-reservoir response rate and molecular-consumption rate are equal, $\gamma_{HI} = \gamma_{H_2}$, then $1/\tau_{conv} + \gamma_{strip} = (1 - R + \eta)/\tau_{dep}$. The integrand in equation (11) is unity, then both H_2 and SFR surface densities can be further reduced to:

$$\begin{aligned} \Sigma_{H_2}(t) &= \left(\Sigma_{H_2,0} + \frac{\Sigma_{HI,0}}{\tau_{conv}} t\right) e^{-\gamma_{H_2} t}, \\ \Sigma_{SFR}(t) &= \frac{1}{\tau_{dep}} \left(\Sigma_{H_2,0} + \frac{\Sigma_{HI,0}}{\tau_{conv}} t\right) e^{-\gamma_{H_2} t}. \end{aligned} \tag{15}$$

3.3 What determines whether the SFR initially rises then decreases or monotonic falls?

Define the initial replenishment time $\tau_{\Phi,0} = \Sigma_{\mathrm{H_2},0}/\Sigma_{\Phi,0}$ and the dimensionless remaining-SFR fraction $F_{\mathrm{SFR}}(t) = \Sigma_{\mathrm{SFR}}(t)/\Sigma_{\mathrm{SFR},0}$. Equation (14) becomes

$$F_{\mathrm{SFR}}(t) = e^{-\gamma_{\mathrm{H_2}} t} + \frac{1}{\tau_{\Phi,0}} \frac{e^{-\gamma_{\mathrm{HI}} t} - e^{-\gamma_{\mathrm{H_2}} t}}{\gamma_{\mathrm{H_2}} - \gamma_{\mathrm{HI}}}. \tag{16}$$

More directly, divide equation (9) by the molecular column and use constant τ_{dep}:

$$\begin{aligned} \frac{1}{\Sigma_{\mathrm{SFR}}(t)} \frac{d\Sigma_{\mathrm{SFR}}(t)}{dt} &= \frac{1}{\Sigma_{\mathrm{H_2}}(t)} \frac{d\Sigma_{\mathrm{H_2}}(t)}{dt} \\ &= \frac{\Sigma_{\Phi}(t)}{\Sigma_{\mathrm{H_2}}(t)} - \gamma_{\mathrm{H_2}} \\ &= \frac{1}{\tau_{\Phi}(t)} - \gamma_{\mathrm{H_2}}. \end{aligned} \tag{17}$$

Thus the sign follows from supply relative to consumption:

$$\begin{aligned} \left.\frac{d\Sigma_{\mathrm{SFR}}(t)}{dt}\right|_0 > 0 \iff \Sigma_{\Phi,0} > \gamma_{\mathrm{H_2}} \Sigma_{\mathrm{H_2},0} \iff \tau_{\Phi,0} < \frac{\tau_{\mathrm{dep}}}{1 - R + \eta}, \\ \left.\frac{d\Sigma_{\mathrm{SFR}}(t)}{dt}\right|_0 < 0 \iff \tau_{\Phi,0} > \frac{\tau_{\mathrm{dep}}}{1 - R + \eta}. \end{aligned} \tag{18}$$

Equality gives zero initial slope, not permanent equilibrium. If the initial molecular reservoir is balanced, substitution of $1/\tau_{\Phi,0} = \gamma_{\mathrm{H_2}}$ gives

$$\begin{aligned} F_{\mathrm{SFR}}(t) &= \frac{\gamma_{\mathrm{HI}} e^{-\gamma_{\mathrm{H_2}} t} - \gamma_{\mathrm{H_2}} e^{-\gamma_{\mathrm{HI}} t}}{\gamma_{\mathrm{HI}} - \gamma_{\mathrm{H_2}}}, \\ \frac{dF_{\mathrm{SFR}}}{dt} &= \frac{\gamma_{\mathrm{HI}} \gamma_{\mathrm{H_2}}}{\gamma_{\mathrm{HI}} - \gamma_{\mathrm{H_2}}} \left(e^{-\gamma_{\mathrm{HI}} t} - e^{-\gamma_{\mathrm{H_2}} t}\right) < 0 \quad (t > 0). \end{aligned} \tag{19}$$

The derivative is negative for either ordering of the positive rates. At the onset, the first derivative vanishes but the curvature is negative:

$$F_{\mathrm{SFR}}(t) = 1 - \frac{\gamma_{\mathrm{HI}} \gamma_{\mathrm{H_2}}}{2} t^2 + O(t^3). \tag{20}$$

This explains the delayed decrease. Atomic removal due to RPS immediately reduces future gas supply, but it does not instantly destroy the molecular reservoir or its ongoing star formation.

At late times the slower exponential usually dominates. In the common regime $\gamma_{\mathrm{HI}} > \gamma_{\mathrm{H_2}}$, the molecular response approaches an exponential with timescale $1/\gamma_{\mathrm{H_2}} = \tau_{\mathrm{dep}}/(1 - R + \eta)$. It is therefore exponential-like, but generally not one exponential from the onset. For any nonnegative supply, equation (13) also implies

$$F_{\mathrm{SFR}}(t) \geq e^{-\gamma_{\mathrm{H_2}} t}. \tag{21}$$

Increasing HI stripping cannot make retained H_2 disappear faster than the no-supply consumption solution (closed-box consumption of H_2).

3.4 A temporal maximum for an initially over-supplied region

Suppose the initial supply exceeds molecular consumption. Differentiating equation (16) and collecting terms gives

$$\frac{dF_{\mathrm{SFR}}}{dt} = \frac{\gamma_{\mathrm{H_2}}[\tau_{\Phi,0}^{-1} + \gamma_{\mathrm{HI}} - \gamma_{\mathrm{H_2}}]e^{-\gamma_{\mathrm{H_2}} t} - \gamma_{\mathrm{HI}}\tau_{\Phi,0}^{-1} e^{-\gamma_{\mathrm{HI}} t}}{\gamma_{\mathrm{H_2}} - \gamma_{\mathrm{HI}}}. \tag{22}$$

At a stationary point t_{peak}, the numerator vanishes. Moving one term to the other side and dividing by the positive exponential gives

$$\begin{aligned} \gamma_{\mathrm{H_2}}[1 + (\gamma_{\mathrm{HI}} - \gamma_{\mathrm{H_2}})\tau_{\Phi,0}]e^{-\gamma_{\mathrm{H_2}} t_{\mathrm{peak}}} &= \gamma_{\mathrm{HI}} e^{-\gamma_{\mathrm{HI}} t_{\mathrm{peak}}}, \\ e^{(\gamma_{\mathrm{HI}} - \gamma_{\mathrm{H_2}}) t_{\mathrm{peak}}} &= \frac{\gamma_{\mathrm{HI}}}{\gamma_{\mathrm{H_2}}[1 + (\gamma_{\mathrm{HI}} - \gamma_{\mathrm{H_2}})\tau_{\Phi,0}]}. \end{aligned} \tag{23}$$

Taking the natural logarithm yields

$$\boxed{t_{\mathrm{peak}} = \frac{1}{\gamma_{\mathrm{HI}} - \gamma_{\mathrm{H_2}}} \ln\left[\frac{\gamma_{\mathrm{HI}}}{\gamma_{\mathrm{H_2}}[1 + (\gamma_{\mathrm{HI}} - \gamma_{\mathrm{H_2}})\tau_{\Phi,0}]}\right]}. \tag{24}$$

This is a positive-time peak of the temporal SFR history, not a measurement of spatial enhancement. It occurs when $\tau_{\Phi,0}^{-1} > \gamma_{\mathrm{H_2}}$. For equal rates, differentiating $(1 + t/\tau_{\Phi,0})e^{-\gamma_{\mathrm{H_2}} t}$ gives the finite limit $t_{\mathrm{peak}} = 1/\gamma_{\mathrm{H_2}} - \tau_{\Phi,0}$.

To verify that the stationary point is a maximum, differentiate the molecular balance once more:

$$\begin{aligned} \frac{d^2\Sigma_{\mathrm{H_2}}(t)}{dt^2} &= -\gamma_{\mathrm{HI}}\Sigma_{\Phi}(t) - \gamma_{\mathrm{H_2}}\frac{d\Sigma_{\mathrm{H_2}}(t)}{dt}, \\ \left.\frac{d^2\Sigma_{\mathrm{H_2}}(t)}{dt^2}\right|_{t_{\mathrm{peak}}} &= -\gamma_{\mathrm{HI}}\Sigma_{\Phi}(t_{\mathrm{peak}}) < 0. \end{aligned} \tag{25}$$

Every stationary point has negative curvature.

Therefore a solution of either Σ_{H_2} or Σ_{SFR} starting with positive slope has one maximum followed by an exponential-like decline; a solution starting with nonpositive slope leads to a monotonic decrease. Finally, we show the indirect route by which atomic deficiency affects molecular gas due to cluster environment, and thus the suppresion of SF (e.g., Fumagalli et al. 2009).

3.5 Atomic-gas asymmetry and the leading/trailing SFR contrast

We now want to explain the potential SF enhancement/suppression in the leading/tailing RPS side, e.g., in NGC4654. Here we compare positions x with different initial atomic columns, but the same initial molecular column $\Sigma_{H_2,0}$ and the same τ_{conv}, τ_{dep}, R, η, and γ_{strip}. Here x denotes a spatial position. The rates in equation (5) are therefore common to all positions. We define $c_{HI}(x) \equiv \Sigma_{HI}(x,0)/\Sigma_{HI,0} > 0$, where $\Sigma_{HI,0}$ is the reference initial atomic column and $c_{HI}(x_{reference}) = 1$.

This factor represents an atomic excess or deficit established before model onset. Applying the perturbation only to HI is again motivated by the greater susceptibility of diffuse atomic gas than dense molecular clouds to ram pressure (Boselli et al. 2014; Lee et al. 2017; Cramer et al. 2020). It is an initial-condition approximation on how RPS act on the compresion of HI gas. Equation (8) and the unchanged supply law immediately give

$$\begin{aligned} \Sigma_{HI}(x,t) &= c_{HI}(x)\Sigma_{HI,0}e^{-\gamma_{HI}t}, \\ \Sigma_{\Phi}(x,t) &= \frac{c_{HI}(x)\Sigma_{HI,0}}{\tau_{conv}}e^{-\gamma_{HI}t}. \end{aligned} \tag{26}$$

The molecular equation remains $d\Sigma_{H_2}(x,t)/dt = \Sigma_{\Phi}(x,t) - \gamma_{H_2}\Sigma_{H_2}(x,t)$. Relative to section 3.2, only the supply amplitude is multiplied by the constant $c_{HI}(x)$. The integrating-factor solution in equation (13) therefore becomes

$$\begin{aligned} \Sigma_{H_2}(x,t) = \Sigma_{H_2,0}e^{-\gamma_{H_2}t} \\ + \frac{c_{HI}(x)\Sigma_{HI,0}}{\tau_{conv}} \frac{e^{-\gamma_{HI}t} - e^{-\gamma_{H_2}t}}{\gamma_{H_2} - \gamma_{HI}}. \end{aligned} \tag{27}$$

Only the supplied term is multiplied by $c_{HI}(x)$; the initial molecular term is common. Dividing by the common depletion timescale gives

$$\begin{aligned} \Sigma_{SFR}(x,t) = \frac{\Sigma_{H_2,0}}{\tau_{dep}}e^{-\gamma_{H_2}t} \\ + \frac{c_{HI}(x)\Sigma_{HI,0}}{\tau_{conv}\tau_{dep}} \frac{e^{-\gamma_{HI}t} - e^{-\gamma_{H_2}t}}{\gamma_{H_2} - \gamma_{HI}}. \end{aligned} \tag{28}$$

For equal response rates, replace the exponential quotient by $te^{-\gamma_{H_2}t}$, as in equation (15).

Let $\Sigma_{\mathrm{H_2}}^{\mathrm{reference}}(t)$ be equation (27) evaluated at $c_{\mathrm{HI}} = 1$, which corresponds to equation (13), i.e., $\Sigma_{\mathrm{H_2}}^{\mathrm{reference}}(t) = \Sigma_{\mathrm{H_2}}(t)$. Subtracting the solutions at any two positions x_1 and x_2 cancels the common initial term:

$$\begin{aligned}&\Sigma_{\mathrm{H_2}}(x_1, t) - \Sigma_{\mathrm{H_2}}(x_2, t)\\ &\quad = \left[c_{\mathrm{HI}}(x_1) - c_{\mathrm{HI}}(x_2)\right]\left[\Sigma_{\mathrm{H_2}}^{\mathrm{reference}}(t) - \Sigma_{\mathrm{H_2},0} e^{-\gamma_{\mathrm{H_2}} t}\right].\end{aligned} \tag{29}$$

The second bracket is the surviving molecular gas supplied after $t = 0$. It is strictly positive for every finite $t > 0$, since the numerator and denominator of the exponential quotient have the same sign; its equal-rate limit is also positive. Consequently, the molecular column difference, and the SFR difference obtained by dividing by $\tau_{\mathrm{dep}} > 0$, have the sign of $c_{\mathrm{HI}}(x_1) - c_{\mathrm{HI}}(x_2)$.

If the proposed geometry establishes $c_{\mathrm{HI}}(x_{\mathrm{leading}}) > 1 = c_{\mathrm{HI}}(x_{\mathrm{reference}}) > c_{\mathrm{HI}}(x_{\mathrm{trailing}}) > 0$, then

$$\boxed{\begin{aligned}&\Sigma_{\mathrm{H_2}}^{\mathrm{leading}}(t) > \Sigma_{\mathrm{H_2}}^{\mathrm{reference}}(t) > \Sigma_{\mathrm{H_2}}^{\mathrm{trailing}}(t),\\ &\Sigma_{\mathrm{SFR}}^{\mathrm{leading}}(t) > \Sigma_{\mathrm{SFR}}^{\mathrm{reference}}(t) > \Sigma_{\mathrm{SFR}}^{\mathrm{trailing}}(t), \qquad t > 0.\end{aligned}} \tag{30}$$

4. From decreasing Σ_{SFR} to fading young-star Hα emission

Let $\mathcal{L}_{\alpha}^{\mathrm{young}}$ be the Hα luminosity per adopted area powered by the young stellar population. It includes both compact HII emission and emission powered by O/B star photons absorbed outside compact HII regions, i.e. the leaky photons. For the main calculation these are one effective component.

The conversion from SFR to an ionizing population normally averages the recent star formation history over the lifetimes of massive stars (Kennicutt & Evans 2012). For gas evolution much slower than that stellar response, we use

$$\boxed{\mathcal{L}_{\alpha}^{\mathrm{young}}(t) \simeq \frac{1}{C_{\alpha}} \Sigma_{\mathrm{SFR}}(t) = \mathcal{L}_{\alpha,0}^{\mathrm{young}} F_{\mathrm{SFR}}(t).} \tag{31}$$

C_{α} is the Hα-to-SFR calibration, which already absorbs stellar population and the assumed IMF. Note that in MAUVE observation the adopted value is $C_{\alpha} = 4.98 \times 10^{-42}\ M_{\odot}\,\mathrm{yr}^{-1}/(\mathrm{erg\,s}^{-1})$ with Chabrier IMF.

5. HOLMES's contribution to Hα budget

Let $\Sigma_*^{\rm old}$ be the current mass surface density in the old population, including its associated remnants, and let $q_{\rm H,HOLMES}$ be the production rate of hydrogen-ionizing photons per unit of that current mass. Their product is the emitted photon production per area:

$$\mathcal{Q}_{\rm HOLMES} = q_{\rm H,HOLMES}\Sigma_*^{\rm old}. \tag{32}$$

Here we adopt $q_{\rm H,HOLMES} = 7 \times 10^{40}\ {\rm s}^{-1} M_\odot^{-1}$ from the PEGASE normalization used by Belfiore et al. (2022), section 3.2. In ionization equilibrium, one absorbed ionizing photon balances a Case-B recombination (Hummer & Storey 1987): α_B excludes recombinations directly to the hydrogen ground state; $\alpha_\alpha^{\rm eff}$ counts recombinations yielding Hα. The probability that such a recombination produces Hα is $p_\alpha = \alpha_\alpha^{\rm eff}/\alpha_B$. Each emitted H$\alpha$ photon carries energy $h\nu_\alpha$. Therefore

$$\begin{aligned} \mathcal{L}_\alpha^{\rm HOLMES} &= h\nu_\alpha p_\alpha \mathcal{Q}_{\rm HOLMES} \\ &= \boxed{\epsilon_\alpha q_{\rm H,HOLMES}\Sigma_*^{\rm old}}, \\ \epsilon_\alpha \equiv h\nu_\alpha p_\alpha, &\qquad p_\alpha \equiv \frac{\alpha_\alpha^{\rm eff}}{\alpha_B} \simeq \frac{1}{2.206}. \end{aligned} \tag{33}$$

With $\lambda_\alpha = 6562.8$Å, $\epsilon_\alpha = 1.3721 \times 10^{-12}$ erg per absorbed photon. The adopted numerical conversion p_α is given in Cid Fernandes et al. (2011), equation 2, for photoionization by populations older than 10^8 yr. Therefore, since the old stellar mass ($\Sigma_*^{\rm old}$) assigned to the region changes little, we consider HOLMES's Hα emission as constant (or at least relatively constant compared to the young stellar's contribution).

6. How the luminosity weights change

By our definition, Hα emission comes from young stellar and HOLMES components, so we can also define their luminosity weights, respectively:

$$\begin{aligned} \mathcal{L}_\alpha(t) &= \mathcal{L}_\alpha^{\rm young}(t) + \mathcal{L}_\alpha^{\rm HOLMES}, \\ w_{\rm HOLMES}(t) &\equiv \frac{\mathcal{L}_\alpha^{\rm HOLMES}}{\mathcal{L}_\alpha^{\rm young}(t) + \mathcal{L}_\alpha^{\rm HOLMES}}, \\ w_{\rm young}(t) &= 1 - w_{\rm HOLMES}(t). \end{aligned} \tag{34}$$

These are light fractions, not fractions of area or gas mass. Write $w_{\rm HOLMES,0}$ $= w_{\rm HOLMES}(0)$. For fixed positive HOLMES luminosity, the quotient rule gives

$$\begin{aligned}\frac{dw_{\rm HOLMES}(t)_{\rm HOLMES}(t)}{dt} &= -\frac{\mathcal{L}_\alpha^{\rm HOLMES}}{[\mathcal{L}_\alpha(t)]^2}\frac{d\mathcal{L}_\alpha^{\rm young}(t)}{dt} \\ &= -w_{\rm HOLMES}(t)[1-w_{\rm HOLMES}(t)]\frac{d\ln\Sigma_{\rm SFR}(t)}{dt} \\ &= w_{\rm HOLMES}(t)[1-w_{\rm HOLMES}(t)]\left[\gamma_{\rm H_2}-\frac{1}{\tau_\Phi(t)}\right].\end{aligned} \tag{35}$$

This connects the gas balance directly to the changing source weight. Molecular consumption exceeding replenishment makes SFR decline and the HOLMES fraction rise. An initially over-supplied region has the opposite response until its SFR maximum. Thus, over the cluster corssing timescale, we expect to see the increase of HOLMES's weight.

7. The BPT line ratios and the complete connection

For a forbidden line ℓ, let B denote its Balmer denominator and define the linear component ratio $R^j_{\ell/B} = \mathcal{L}^j_\ell/\mathcal{L}^j_B$, with j belongs to young or HOLMES contribution. N2 is [N II]6583/Halpha, S2 is ([S II]6716+[S II]6731)/Halpha, and O3 is [O III]5007/Hbeta. Due to the harder ionization of HOLMES, we have

$$R^{\rm young}_{\ell/B} < R^{\rm HOLMES}_{\ell/B}. \tag{36}$$

For simplicity, these two terms are set to be constant to indicate the fact that these two sources have representative BPT line ratios; in reality, temperature, ionic fractions, metallicity, N/O, and ionization parameter can change the line ratios (Byler et al. 2019).

We adopt the intrinsic Balmer ratio $\mathcal{L}^j_\alpha/\mathcal{L}^j_\beta = 2.86$ for each component, an approximation consistent with the Case-B convention used for the observational dust correction (Hummer & Storey 1987). Thus

$$\frac{\mathcal{L}^{\rm HOLMES}_\beta}{\mathcal{L}_\beta} = \frac{\mathcal{L}^{\rm HOLMES}_\alpha/2.86}{\mathcal{L}_\alpha/2.86} = w_{\rm HOLMES}. \tag{37}$$

Substitute $\mathcal{L}^j_\ell = R^j_{\ell/B}\mathcal{L}^j_B$ into the total ratio yields the evolution of BPT line ratio, which show the similar forms of equations 7 and 8 in Blanc et al. (2009):

$$
\begin{aligned}
R_{\ell/B}(t) &= \frac{\mathcal{L}_l^{\rm young}(t) + \mathcal{L}_l^{\rm HOLMES}}{\mathcal{L}_B^{\rm young}(t) + \mathcal{L}_B^{\rm HOLMES}} \\
&= \frac{R_{\ell/B}^{\rm young}\mathcal{L}_B^{\rm young}(t) + R_{\ell/B}^{\rm HOLMES}\mathcal{L}_B^{\rm HOLMES}}{\mathcal{L}_B^{\rm young}(t) + \mathcal{L}_B^{\rm HOLMES}} \\
&= [1 - w_{\rm HOLMES}(t)]R_{\ell/B}^{\rm young} + w_{\rm HOLMES}(t)R_{\ell/B}^{\rm HOLMES} \\
&= \boxed{R_{\ell/B}^{\rm young} + (R_{\ell/B}^{\rm HOLMES} - R_{\ell/B}^{\rm young})w_{\rm HOLMES}(t)}.
\end{aligned}
\tag{38}
$$

Then we differentiate equation (38):

$$
\frac{dR_{\ell/B}(t)}{dt} = (R_{\ell/B}^{\rm HOLMES} - R_{\ell/B}^{\rm young})\frac{dw_{\rm HOLMES}(t)_{\rm HOLMES}(t)}{dt}.
\tag{39}
$$

Hence, the change of HOLMES weight/fraction directly trace the change of observed BPT line ratio.

Finally, inserting the gas solution, the young-star conversion, and the old-star normalization gives the complete prediction:

$$
\boxed{
\begin{aligned}
\mathcal{L}_\alpha(t) &= \frac{\Sigma_{\rm SFR,0}}{C_\alpha}F_{\rm SFR}(t) + \epsilon_\alpha q_{\rm H,HOLMES}\Sigma_*^{\rm old}, \\
R_{\ell/B}(t) &= R_{\ell/B}^{\rm young} + (R_{\ell/B}^{\rm HOLMES} - R_{\ell/B}^{\rm young})\frac{\epsilon_\alpha q_{\rm H,HOLMES}\Sigma_*^{\rm old}}{\mathcal{L}_\alpha(t)}.
\end{aligned}
}
\tag{40}
$$

8. Extension: distinguishing H II and leaked OB-star radiation

Motivated by the spectral hardening of leaked OB radiation discussed by Baron et al. (2026), to further distinguish the the two young-star-powered components, we introduce the time-dependent escape fraction $f_{\rm esc}(t)$ of OB-star ionizing photons originated from compact H II regions, while retaining the original gas and Hα luminosity evolution.

For this simple extension, we identify $f_{\rm esc}(t)$ with the fraction of *young-star-powered Balmer emission* attributed to leaked photons, assuming these photons are subsequently absorbed within the modeled region. Define constant intrinsic line ratios $R_{\ell/B}^{\rm HII}$ and $R_{\ell/B}^{\rm leak}$ for the compact H II and leaked-radiation components, respectively. Then

$$
\boxed{R_{\ell/B}^{\rm young}(t) = [1 - f_{\rm esc}(t)]R_{\ell/B}^{\rm HII} + f_{\rm esc}(t)R_{\ell/B}^{\rm leak}.}
\tag{41}
$$

Then we can expand the equation (38), so the observed ratio becomes

$$
\begin{aligned}
R_{\ell/B}(t) &= [1 - w_{\rm HOLMES}(t)]([1 - f_{\rm esc}(t)]R_{\ell/B}^{\rm HII} + f_{\rm esc}(t)R_{\ell/B}^{\rm leak}) \\
&\quad + w_{\rm HOLMES}(t)R_{\ell/B}^{\rm HOLMES} \\
&= [1 - w_{\rm HOLMES}(t)][1 - f_{\rm esc}(t)]R_{\rm HII} \\
&\quad + [1 - w_{\rm HOLMES}(t)]f_{\rm esc}(t)R_{\rm leak} \\
&\quad + w_{\rm HOLMES}(t)R_{\rm HOLMES},
\end{aligned}
\tag{42}
$$

where the ℓ/B subscripts on the component ratios are omitted for compactness. As in Section 7, we adopt a common intrinsic Balmer decrement for the three components. We assume that the intrinsic component ratios remain constant and, for the line ratio being considered, satisfy

$$
R_{\rm HII} < R_{\rm leak}, \qquad R_{\rm HII} < R_{\rm HOLMES}, \qquad R_{\rm young}(t) < R_{\rm HOLMES}. \tag{43}
$$

Crucially, **no particular ordering between $R_{\rm leak}$ and $R_{\rm HOLMES}$ is required**. The last inequality is the condition for the growing HOLMES weight to increase the observed ratio; it must remain valid over the modeled interval.

RPS lowers the available neutral gas column, potentially opening lower-opacity channels for escaping ionizing photons (Boselli et al. 2022; Gazagnes et al. 2018). We therefore adopt $df_{\rm esc}/dt > 0$ as a plausible simplifying assumption after stripping, rather than a universal empirical result. During the fading-SFR phase, equation (35) gives $dw_{\rm HOLMES}(t)/dt > 0$. Differentiating equation (42) then yields

$$
\boxed{
\begin{aligned}
\frac{dR_{\rm obs}}{dt} &= \underbrace{\frac{dw_{\rm HOLMES}(t)}{dt}(R_{\rm HOLMES} - R_{\rm young})}_{>0\ \text{(HOLMES)}} \\
&\quad + \underbrace{(1-w)\frac{df_{\rm esc}}{dt}(R_{\rm leak} - R_{\rm HII})}_{>0\ \text{(leakage)}} \\
&> 0.
\end{aligned}
}
\tag{44}
$$

Thus, under these extended conditions and assumptions, the observed forbidden-line-to-Balmer ratio still increases as star formation fades: both the rising *relative HOLMES contribution* and the increasing *fraction of leaked OB-star radiation* contribute positively. When $df_{\rm esc}/dt = 0$, the extension reduces to the original result in equation (39). This provides a simple combined interpretation of the leakage-driven hardening emphasized by Baron et al. (2026) and the HOLMES contribution explored by Belfiore et al. (2022).

9. Conclusion

In summary, by establishing the simple two-reservoir model, we show that as a galaxy goes through the cluster environment, the RPS acts on the removal of HI column and cause the delayed decline of both molecular gas and SFR intensity. This further leads to the fading Hα surface brightness from young stellar component, while the Hα emission from ionization by HOLMES remains relatively unchanged, which causes the increasing fraction/weight of harder ionization and therefore higher BPT line ratio in observation.

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